Proposal: Enhancing Credit Risk Assessment via Machine Learning

To: Team Lead

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Subject: Transitioning from Manual Review to Predictive Risk Modelling

1. The Objective

Currently, our credit risk process is manual, making it slow and vulnerable to human oversight. I propose integrating a Machine Learning (ML) Model into our pipeline. This will transform our "Under Review" stage from a manual bottleneck into an automated, data-driven engine.

2. How the System Works

Instead of a person reading every line of a bank statement, the system will act as a Smart Filter:

• Data Ingestion: The model looks at a customer’s financial history (income, previous debt, and payment consistency).

• Pattern Recognition: It compares the current applicant against thousands of past successful and failed loans to find hidden risk patterns that a human eye might miss.

• The Risk Score: The system generates a "Confidence Score." High-score applications are fast-tracked for approval, while low-score ones are flagged for a closer look.

3. Key Benefits

• Speed: Reduces the decision time from days to seconds.

• Consistency: The AI applies the same rules to every applicant, removing personal bias or "fatigue" from manual reviewers.

• Scalability: We can process 1,000 applications as easily as 10.

4. Potential Challenges & Solutions

• "Black Box" Problem: We can’t just let the AI say "No" without a reason.

• Solution: We will use Explainable AI tools so the system can tell us exactly why a loan was flagged (e.g., "Debt-to-income ratio too high").

• Data Quality: If the input data is old, the prediction will be wrong.

• Solution: We will implement regular "Model Retraining" to keep the AI updated with current market trends.

• Security: We must protect the applicant's data during the AI processing phase.

• Solution: All data used for training will be encrypted and anonymized.

5. Conclusion

Adding an ML step doesn’t mean we replace humans; it means we give our team better tools to make faster, safer decisions. I recommend we start with a Pilot Phase using historical data to test the model's accuracy before going live.